Quantum system in state $|\psi\rangle$ $\boxed{\text{Linear Response}}$

$$O = \langle\psi|O|\psi\rangle$$

$$|\psi\rangle = |\psi(t)\rangle \quad \text{via} \quad i\hbar\,\partial_t|\psi(t)\rangle = \underbrace{H}_{t\text{-independent}}|\psi(t)\rangle$$

$$\Rightarrow \quad O(t) = \langle\psi(t)|O|\psi(t)\rangle$$

But $|\psi(t)\rangle = e^{-iHt/\hbar}|\psi(t=0)\rangle$

$$\Rightarrow \quad O(t) = \langle\psi(t)|\,O\,|\psi(t)\rangle =$$

$$= \langle\psi(t=0)|\,e^{iHt/\hbar}\,O\,e^{-iHt/\hbar}\,|\psi(t=0)\rangle$$

Heisenberg representation

operators depend on time, wave functions – not

$$O_H(t) = e^{iHt/\hbar}\,O\,e^{-iHt/\hbar}$$

(Schrodinger representation: $\psi(t)$ depends on time, operators – not)

$$\Rightarrow \quad O(t) = \langle\psi_0|\,O_H(t)\,|\psi_0\rangle$$

$\psi_0 = \psi(t=0)$

$$i\hbar\frac{d}{dt}O_H(t) = i\hbar\left\{\frac{i}{\hbar}H e^{iHt}\,O\,e^{-iHt/\hbar}\right.$$

$$\left. + e^{iHt/\hbar}\,O\,e^{-iHt/\hbar}\left(-\frac{iH}{\hbar}\right)\right\} = -H O_H + O_H H$$

$$\Rightarrow \frac{d}{dt} O_H(t) = \frac{i}{\hbar} [H, O_H(t)]$$

For a system not in a pure state

$$\langle O \rangle = \sum_n p_n \langle \psi_n | O | \psi_n \rangle$$

let $\hat{\rho} = \sum_n p_n |\psi_n\rangle\langle\psi_n|$

then $\langle O \rangle = Tr[\hat{\rho}\, O] =$

$$= \sum_n p_n \, Tr[|\psi_n\rangle\langle\psi_n| O]$$

$$= Tr[O |\psi_n\rangle\langle\psi_n|]$$

$$= Tr[\langle\psi_n| O |\psi_n\rangle]$$

Linear response: H_0 t-independent (at $t=-\infty$)

at some time time slowly turn perturbation $H_{pert}(t) = H'(t) = \int dr\, h(\vec{r},t)\, O(r)$

external field

Then $H(t) = H_0 + H'(t)$ is t-dependent

$$\langle A \rangle_t = Tr[\rho(t) A]$$

Choose A such that $Tr[\rho_0 A] = 0$

(for example, A can represent current and $H'(t)$ describes external electric field $\Rightarrow$ then the current appears as a response to el. field)

Adiabatic evolution

$H(t) = H_0 + H'(t)$, where $H'(t \to -\infty) = 0$.

At $t = -\infty$ the system is described by an equilibrium statistical operator

$$\rho_0 = \frac{1}{Z} \sum_{n=0}^{\infty} e^{-\frac{E_n}{T}} |n\rangle\langle n| = \frac{1}{Z} \sum_n e^{-E_n/T} |\psi_n\rangle\langle\psi_n|$$

where E_n are eigenvalues of H_0, and ψ_n are its eigenvectors.

When $H'(t)$ switches on, the wave functions ψ_n start to evolve, $\psi_n \to \psi_n(t)$. At this point we <u>assume</u> that evolution is <u>adiabatic</u>, which means that probability p_n to occupy level <u>n</u> are <u>not</u> changing with time, $p_n = \frac{e^{-E_n/T}}{Z}$ is t-independent.

If so, then $i\hbar \frac{\partial}{\partial t} \psi_n = H(t)\, \psi_n(t)$

Since now $\rho(t) = \sum_n \frac{e^{-E_n/T}}{Z} |\psi_n(t)\rangle\langle\psi_n(t)|$, we find

$$i\hbar \frac{\partial}{\partial t} \rho(t) = \sum_n p_n \left(i\hbar \frac{\partial}{\partial t} |\psi_n(t)\rangle\langle\psi_n(t)| + \right.$$

$$\left. + |\psi_n(t)\rangle\langle i\hbar \frac{\partial}{\partial t} \psi_n(t)| \right)$$

$$i\hbar \frac{\partial}{\partial t} |\psi_n(t)\rangle = \hat{H}(t)\, |\psi_n(t)\rangle$$

$$-i\hbar \frac{\partial}{\partial t} \langle\psi_n(t)| = \langle\psi_n(t)|\, \hat{H}(t)$$

$$\Rightarrow i\hbar \frac{\partial}{\partial t} \rho(t) = \sum_n p_n \left(\hat{H}(t)\, |\psi_n(t)\rangle\langle\psi_n(t)| - |\psi_n(t)\rangle\langle\psi_n(t)|\, \hat{H}(t) \right) =$$

$$= [\hat{H}(t), \rho(t)]$$

To obtain <u>linear response</u>, we parameterize

$$\rho(t) = \rho_0 + f(t) \quad , \text{where } f(t) \sim \hat{H}'(t)$$

and ρ_0 is equilibrium density matrix at $t=-\infty$.

$$\Rightarrow i\hbar \frac{\partial}{\partial t}\left(\rho_0 + f(t)\right) = \left[\hat{H}_0 + \hat{H}'(t), \rho_0 + f(t)\right] =$$

$$= \underbrace{\left[\hat{H}_0, \rho_0\right]}_{=0} + \left[\hat{H}_0, f(t)\right] + \left[\hat{H}'(t), \rho_0\right] + \underbrace{\left[\hat{H}'(t), f(t)\right]}_{\text{neglect because this is } \sim (\hat{H}')^2}$$

because $\rho_0 = \dfrac{e^{-\hat{H}_0/T}}{Z}$

Thus $i\hbar \dfrac{\partial}{\partial t} f(t) - \left[H_0, f(t)\right] = \left[H'(t), \rho_0\right]$

~~This~~ The left-hand side can be re-written as

$$e^{-iH_0t/\hbar}\left\{ i\hbar \frac{\partial}{\partial t}\left(e^{\frac{i}{\hbar}H_0t} f(t)\, e^{-\frac{i}{\hbar}H_0t}\right)\right\} e^{\frac{i}{\hbar}H_0t} =$$

$$= i\hbar \frac{\partial}{\partial t} f(t) - \left[H_0, f(t)\right]$$

Indeed, Consider

$$e^{-\frac{i}{\hbar}H_0 t}\left[i\hbar\,\partial_t\left(e^{\frac{i}{\hbar}H_0 t} f(t)\, e^{-\frac{i}{\hbar}H_0 t}\right)\right] e^{\frac{i}{\hbar}H_0 t} =$$

$$= e^{-\frac{i}{\hbar}H_0 t}\; i\hbar\left(\frac{iH_0}{\hbar}\, e^{\frac{i}{\hbar}H_0 t} f\, e^{-\frac{i}{\hbar}H_0 t} + e^{\frac{i}{\hbar}H_0 t}\,\partial_t f\, e^{-\frac{i}{\hbar}H_0 t}\right.$$

$$\left. + e^{\frac{i}{\hbar}H_0 t} f(t)\, e^{-\frac{i}{\hbar}H_0 t}\left(\frac{-iH_0}{\hbar}\right)\right) e^{\frac{i}{\hbar}H_0 t} =$$

$$= -H_0 f(t) + i\hbar\,\partial_t f + f(t) H_0 =$$

$$= -[H_0, f] + [H_0, f] + [H', \rho_0]$$

$$\Rightarrow \quad i\hbar\frac{\partial}{\partial t}\left(e^{iH_0 t/\hbar} f(t)\, e^{-iH_0 t/\hbar}\right) = e^{\frac{i}{\hbar}H_0 t}[H', \rho_0]\, e^{-\frac{i}{\hbar}H_0 t}$$

$$= \Big[\underbrace{e^{i\frac{H_0 t}{\hbar}} H' e^{-i\frac{H_0 t}{\hbar}}}_{H'_H(t)}, \rho_0\Big]$$

$H'_H(t)$ = perturbing H' in Heisenberg representation

$$\Rightarrow \quad e^{iH_0 t/\hbar} f(t)\, e^{-iH_0 t/\hbar} = -\frac{i}{\hbar}\int_{-\infty}^{t} dt'\, [H'_H(t'), \rho_0]$$

$$f(t) = -\frac{i}{\hbar}\, e^{-iH_0 t/\hbar}\int_{-\infty}^{t} dt'\, [H'_H(t'), \rho_0]\, e^{iH_0 t/\hbar}$$

Evolution of observable A:

$$\langle A \rangle_t = \mathrm{Tr}[\rho(t) A] = \mathrm{Tr}[(\rho_0 + f(t)) A]$$

$$= \underbrace{\mathrm{Tr}[\rho_0 A]}_{\substack{\shortparallel \\ 0}} + \mathrm{Tr}[f(t) A]$$

because $\langle A \rangle = 0$ in the absence of perturbation.
(by assumption)

$$\Rightarrow \langle A \rangle_t = -\frac{i}{\hbar} \mathrm{Tr}\Big\{ \int_{-\infty}^{t} dt' \, e^{-iH_0 t'/\hbar} \cdot [e^{iH_0 t'/\hbar} H'(t') \cdot$$

$$\cdot \, e^{-iH_0 t'/\hbar}, \rho_0] \, e^{iH_0 t/\hbar} A \Big\}$$

move this here using trace properties
$\Downarrow$
get $A(t)$

$$H'(t) = \int d\vec{r} \, h(\vec{r}, t) \, O(\vec{r})$$

$$\langle A \rangle_t = -\frac{i}{\hbar} \mathrm{Tr}\Big\{ \int_{-\infty}^{t} dt' \Big[e^{-iH_0 t'/\hbar} \int d\vec{r} \, h(\vec{r}, t') \, O(\vec{r}) \, e^{-iH_0 t'/\hbar}, \rho_0 \Big]$$

$$\cdot \, A(t) \Big\}$$

$$= -\frac{i}{\hbar} \int_{-\infty}^{t} dt' \int d\vec{r} \, h(\vec{r}, t') \, \mathrm{Tr}\{ [O(r, t'), \rho_0] \, A(t) \}$$

$$\mathrm{Tr}\{[A,B]C\} = \mathrm{Tr}\{ABC - BAC\} =$$

$$= \mathrm{Tr}\{ABC\} - \mathrm{Tr}\{BAC\} = \mathrm{Tr}\{B(CA-AC)\}$$

$$\| $$

$$\mathrm{Tr}\{BCA\}$$

$$= \mathrm{Tr}\{B[C,A]\}$$

$$\Rightarrow \langle A(\vec{r})\rangle_t = -\frac{i}{\hbar}\int_{-\infty}^{t} dt' \int d\vec{r}\,' \, h(\vec{r}\,',t')\, \mathrm{Tr}\{\rho_0 [A(r,t), O(r',t')]\}$$

$$= -\frac{i}{\hbar}\int_{-\infty}^{t} dt' \int d\vec{r}\,' \, \langle [A(r,t), O(r',t')]\rangle_0 \, h(\vec{r}\,',t')$$

$$\langle X\rangle_0 = \mathrm{Tr}\{\rho_0 X\}$$ average with density matrix of H_0

Time-dependence of operators A and O is given by Heisenberg representation

$$A(t) = e^{iH_0 t/\hbar} A\, e^{-iH_0 t/\hbar}$$

$$A(r,t) = \langle A(r)\rangle_t = \int d\vec{r}\,' dt' \, \underbrace{\chi(\vec{r},\vec{r}\,';t,t')}_{\text{susceptibility}} \, h(\vec{r}\,',t')$$

Thus $\chi(\vec{r},\vec{r}\,';t,t') = -\frac{i}{\hbar}\,\underbrace{\theta(t-t')}\,\langle[A(\vec{r},t),O(\vec{r}\,',t')]\rangle_0$

reflects causality:
response occurs at times
<u>after</u> the perturbation
(retarded correlation function)

$\hat{H}_0$ is t-independent and χ is expressed in terms of operators which only know of H_0

$\Rightarrow$ <u>linear response</u> probes properties of H_0.

$$A(t) = \int dt' \, \chi(t,t') \, h(t')$$

$$\underbrace{\int dt \, e^{i\omega t} A(t)}_{A(\omega)} = \int dt \, dt' \, \chi(t,t') \, h(t') \, e^{i\omega t}$$

$$A(\omega) = \int dt \, dt' \, \chi(t,t') \, h(t') \, e^{i\omega t}$$

But χ is determined by unperturbed Hamiltonian which is t-independent $\Rightarrow \chi(t,t') = \chi(t-t')$.

So let $\chi(\Omega) = \int d\tau \, e^{i\Omega\tau} \chi(\tau)$

$$\chi(\tau) = \int \frac{d\omega}{2\pi} \, \chi(\omega) \, e^{-i\omega\tau}$$

$$A(\omega) = \int dt \, dt' \underbrace{\int \frac{d\Omega}{2\pi} \, \chi(\Omega) \, e^{-i\Omega(t-t')}}_{\chi(t-t')} \, e^{i\omega t} \, h(t')$$

$$= \int \frac{d\Omega}{2\pi} \underbrace{\int dt \, e^{it(\omega-\Omega)}}_{2\pi\,\delta(\omega-\Omega)} \underbrace{\int dt' \, e^{i\Omega t'} \, h(t')}_{h(\omega)} \, \chi(\Omega)$$

$$\Rightarrow A(\omega) = \int d\Omega \, \delta(\omega-\Omega) \, \chi(\Omega) \, h(\omega) = \chi(\omega) \, h(\omega)$$

* Response is diagonal in frequency (and momentum) space

$$A(\vec{q},\omega) = \chi(\vec{q},\omega) \, h(\vec{q},\omega)$$

$$\chi(\vec{q},\omega) = \int d\vec{r} \, dt \, e^{-i(\vec{q}\cdot\vec{r} - \omega t)} \, \chi(\vec{r},t)$$

But $\chi(\vec{r},\vec{r}';t,t') = -i\theta(t-t')\langle[\hat{A}(\vec{r},t),\hat{O}(\vec{r}',t')]\rangle_0$

and perturbation $h(\vec{r}',t')$ <u>vanishes</u> at time $t'\to-\infty$.

Let us write it as $h(\vec{r}',t')\,e^{\delta t'}$ with $\underline{\delta=+0}$

so that $e^{\delta t'} \xrightarrow[t\to-\infty]{} 0$ and $h(r;t')$ is otherwise arbitrary (finite) function of time

Then $A(t) = \int dt'\,\chi(t,t')\,h(t')\,e^{\delta t'} =$

(use time invariance of χ) $= \int dt'\,\underbrace{\chi(t-t')\,e^{\delta(t'-t)}}\,e^{\delta t}\,h(t')$

$$= e^{\delta t}\int dt'\,\chi(t-t')\,e^{-(t-t')\delta}\,h(t').$$

We see that $A(t\to-\infty) \sim e^{\delta t} = 0$, as it should be.

Thus $\chi(\omega) = \int d\tau\, e^{i\omega\tau}\,\underbrace{e^{-\delta\tau}\,\chi(\tau)}$

Finally $\chi(\omega) = \int d\tau\, e^{i\omega\tau-\delta\tau}\left(-i\,\theta(\tau)\langle[\hat{A}(\tau),\hat{O}(0)]\rangle_0\right)$

$$= -i\int_0^\infty dt\, e^{i(\omega+i\delta)t}\langle[A(t),O(0)]\rangle_0$$

Retarded function: frequency contains $i\delta = +i0$ part.